I. EXPERIMENTAL METHODS

The experiments were conducted on the clean Bi-terminated surface of β-$Bi_2Pd(001)$ samples, grown following the procedure described in [1]. The crystals were cleaved at approximately 100 K under ultra-high vacuum (UHV) conditions (P<10^{-10} mbar) just before being transferred to a low-temperature scanning tunneling microscope (JT-SPECS GmbH) with a base temperature of 1.3 K. The $\beta-Bi_2Pd$ substrate becomes superconducting below a critical temperature of 4.5 K, exhibiting a superconducting gap of $\Delta_{sample} = 0.775$ meV at this temperature.

All measurements were performed using a $\beta-Bi_2Pd$ superconducting tip. A tungsten (W) tip was in situ coated with superconducting substrate material by repeated indentations onto the clean $\beta-Bi_2Pd(001)$ surface until the width of the superconducting gap in the tunneling spectra doubled. This confirmed the formation of a symmetric Superconductor-Insulator-Superconductor (SIS) junction between the tip and the sample, enabling an energy resolution below 0.06 meV in the spectra. In an SIS junction, the spectroscopic features of the substrate are shifted by the superconducting gap of the tip, which in our results is $\Delta_{tip} \sim 0.775$ meV (see below).

Vanadium adatoms, sourced from a 99.8 % purity rod (Goodfellow Inc.), were deposited directly onto the sample in the STM stage at temperatures below 15 K. Several coverage levels were explored throughout the experiment to rule out coupling effects between magnetic impurities [2].

II. QUASIPARTICLE INTERFERENCE OF $\beta-Bi_2Pd$

To study quasiparticle interference (QPI) in the main manuscript, we perform Fast Fourier Transformation (FFT) of dI/dV images of the $\beta-Bi_2Pd$ sample obtained at a fixed bias voltage value applied to the substrate of the STM. Using bias values to tunnel well above the superconducting gap allows us to explore the QPI of electronic bands in the normal metal density of states with all kinds of scattering elements at the surface. Using sub-gap voltage bias, we explore the scattering of Bogoliubov quasiparticles bound to magnetic impurities exclusively, thus involving only bands hybridized with the magnetic impurity.

Normal state QPI: The quasiparticle interference (QPI) patterns outside the superconducting gap of $\beta-Bi_2Pd$ were previously investigated in Ref. [3]. In our study, we performed large-area constant-current dI/dV mapping on the bare substrate, as shown in Fig. S1a. The presence of surface and subsurface vacancies, along with bismuth adatoms, induces local density of states (LDoS) modulations across the surface, which are captured in maps such as those in Figs. S1b and S1c. Since no magnetic impurities are present, the observed QPI arises from finite potential scattering caused by defects and bismuth adatoms.

The Fast Fourier Transform (FFT) of these dI/dV maps reveals four-fold symmetric contours, consistent with the band structure of $\beta-Bi_2Pd$. To enhance weak signals, we applied a four-fold symmetrization procedure by averaging the maps over four 90° rotations. This approach results in FFT maps such as the one shown in Fig. S2a. Additionally, the resulting scattering vectors can be extracted from line cuts of the FFT intensity, as illustrated in Fig. S2b.